

Redefining PropTech Platform

This case study demonstrates the strategic design approach and leadership applied to revamp a legacy PropTech application, addressing critical pain points in information architecture and user experience through spatial-first principles.

ROLE

Principal Design Consultant

PLATFORM

Web App



Executive Summary

The Challenge

Legacy application suffered from fragmented information architecture and high-friction user experience, relying on static lists disconnected from physical context.

Approach

Introduced a "Spatial-First" discovery model centered on map-integrated data and immersive visualization to bridge the gap between abstract listings and physical reality.

Outcome

Transformed the application into an immersive discovery platform, significantly improving lead quality and user trust through verified, context-rich data visualization.



The Challenge

LEGACY SYSTEM ANALYSIS

Fragmented Information Architecture

Users relied on static images and text-based listings, making it impossible to perceive the true spatial context, scale, and surrounding environment of properties. The lack of spatial visualization created disconnect between user expectations and ground reality.

Filter Fatigue & Cognitive Overload

Generic filtering systems forced users to navigate irrelevant options, leading to decision paralysis. The existing filter architecture was not suited to the needs of different user personas, from individual investors to large-scale commercial developers.

Trust Gap

A significant disconnect existed between curated online presentation and actual physical conditions. Buyers and investors hesitated without verified legal documentation and real-time imagery available upfront, creating friction in the sales funnel.

Core Design Principles

The Trust Gap

Users prioritize "ground reality" - the physical surroundings, access points, and neighborhood context - over abstract metrics. The map became the primary sense-making tool.

Transparency Builds Trust

Easy access to verified legal documentation and high-quality imagery (360° views) is the primary driver for qualified leads. Trust tags and verification status are prominently surfaced.

Smart Filtering, Not Generic Filtering

Filters were redesigned to reduce cognitive load by organizing them into clear, domain-specific buckets that differ based on user persona and property type.

Discovery & Key Insights

USER RESEARCH

11 Users

Investors

Buyers (End Use)

Commercial tenant

Land Buyer/Developer

NRI / Overseas buyer

Context is Critical

"Show me the neighborhood. Is there a road, a park, or industrial runoff next door?"

The Insight: Users prioritize "ground reality" over abstract data. The physical surroundings significantly impact the value perception of land assets.

Filter Fatigue

"I spend more time unchecking boxes than actually finding what I need."

The Insight: Existing generic filters don't apply well to large-scale plots. Users get overwhelmed by options that aren't relevant to industrial needs.

Verification is King

"If I can't see the legal docs upfront, I assume something is wrong."

The Insight: Transparency creates trust. Easy access to verified documentation and real-time imagery is the primary driver for high-quality leads.

Sample User journeys

PROPERTY DISCOVERY PLATFORM

User Journey

Maps

End-to-end experience maps for three distinct intent personas — Renter, Buyer, and Investor — tracking actions, emotions, touchpoints, and platform opportunities.

Renter

Buyer (End-Use)

Investor

NRI / Overseas Buyer

PERSONA

The Investor

Vikram, 42 · Serial Entrepreneur
Deploying ₹50L into a yield-generating asset

GOAL

Deploy ₹50L in a property that generates gross yield AND 10%+ CAGR appreciation. horizon.

1 · THESIS

2 · MARKET SCAN

3 · FILTER

ACTIONS

Reviews investment portfolio. Decides to allocate ₹50L to real estate for diversification + rental yield.

Reads market reports. Identifies Hyderabad / Pune as emerging markets. Looks for micro-market data.

Activates Investor mode. Sets filters: yield >4%, budget ₹50L, 2BHK, emerging localities. Searches. 40+ results.

TOUCHPOINTS

Portfolio review

CA / Wealth advisor

Platform Market Reports

Micro-market Spotlights

Zone Heatmaps

Investor Mode UI

Yield Filter

Cap Rate

Risk Profile Slider

"I want 4%+ yield AND capital upside. Is that possible?"

"Hyderabad looks promising but which micro-market? Looking for rental yield + growth."

"Finally — yield shown on the card. No more chasing dreams."

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Commercial Tenant

Land Buyer / Developer

PERSONA

Land Buyer / Developer

Two sub-types: Suresh, 50 · HNI land banker + Ananya Developers · Mid-size township developer

GOAL

Acquire a plot / agricultural land / NA land along a high-appreciation corridor. Hold 3–7 yrs (banker) OR develop into plotted / mixed-use immediately (developer). ₹50L–₹25Cr range.

MOTIVATIONS

Infrastructure-led appreciation · Convertibility (agri → NA) · Clean title · Low entry cost vs built-up · Development margin (developer sub-type) · RERA-approved layout (plotted)

FEARS

Title disputes · Encroachments · Illegal plots · NA conversion uncertainty · Flood / soil issues · Distant from growth corridor · Developer fraud in plotted schemes

1 · THESIS

2 · CORRIDOR RESEARCH

3 · DISCOVERY

4 · LEGAL DILIGENCE

5 · GROUND INSPECTION

6 · NEGOTIATE & BUY

7 · HOLD / DEVELOP

ACTIONS

Suresh (banker): decides to move ₹1.5Cr from FDs into land near Pune ring road. Ananya Dev: identifies greenfield zone for 200+ plot township project near NH-48.

Studies infrastructure news: upcoming ring roads, DMIC corridor, highway bypasses, SEZ announcements, IT park expansions. Maps growth vectors on paper.

Searches platform in Land mode: "NA plot 2 acres NH-48 within 5km ring road Pune under 60L per acre". Filters: zone type, FSI, road width, CLU status, title clarity score.

Engages title verification lawyer. Checks encumbrance certificate (EC), 7/12 extract, RTC records, mutation entries, and zone certificate from local planning authority.

Visits site physically. Checks actual boundary markers vs map, road access quality, surrounding land use, soil type, water table, neighbouring encroachments.

Negotiates on a per-acre rate benchmarked to recent registrations in taluka. Pays stamp duty on guideline value vs agreed value. Registers in sub-registrar office.

Suresh: holds and monitors — tracks corridor infra progress, checks price appreciation. Ananya: initiates layout approval, RERA registration, construction, sales launch.

TOUCHPOINTS

Economic Times

NHAI project list

Platform Infrastructure Map

Corridor Heatmap

Platform Land Mode

Zone / FSI Filter

EC / 7-12 Extract Guide

Title Lawyer Network

Satellite / Aerial View

Soil Report (if available)

Guideline Value Tool

Recent Registrations Data

Land Portfolio Tracker

PRINT / EXPORT PDF

Infra Pipeline Alerts

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Land Buyer / Developer

PERSONA

Commercial Tenant

GOAL

Lease 3,000–4,000 sq ft Grade A office in Lower Parel / BKC for 3 years. Budget ₹1.8L/month all-in. Needs to attract talent.

MOTIVATIONS

Talent attraction · Brand image · Metro access for employees · Fit-out flexibility · Predictable all-in cost

FEARS

Hidden CAM charges · Poor transit access · Being locked in

4 · SITE VISITS

5 · COMPARE

6 · NEGOTIATE

4 shortlisted spaces. Head of People and HR Lead. Photos, measurements, ceiling height checks.

Compares 3 final options on platform: all-in cost, CAM charges, floor plate efficiency, transit score for employees.

Negotiates fit-out allowance, rent-free (fit-out time), and break clause at 18 months. Engages commercial lawyer.

Scheduler

Side-by-Side Compare

Market Rent Benchmarking Factsheet (PDP)

CAM Breakdown

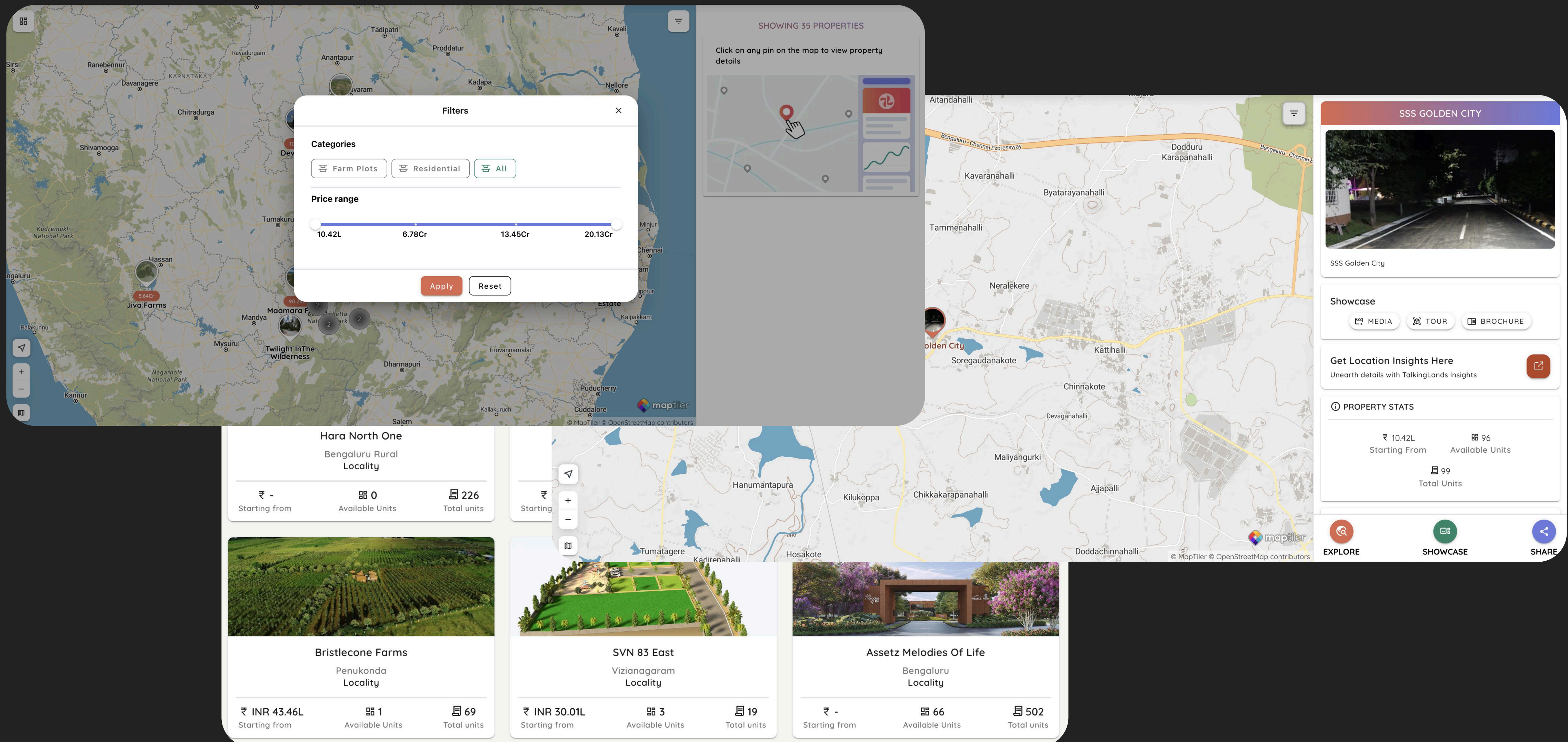
Comparable Leases

Plan

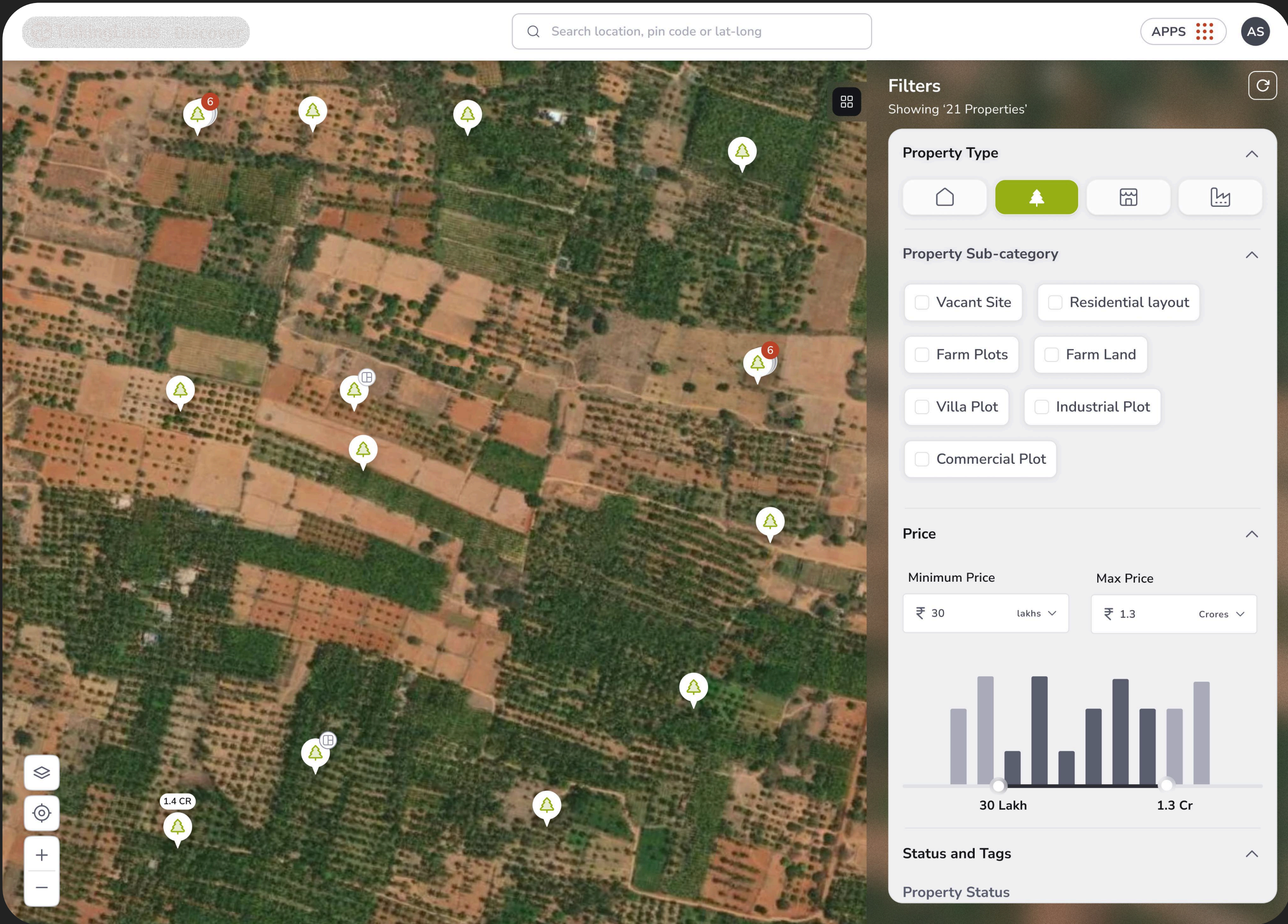
Employee Commute Score

Commercial Lawyer

Current Application



The Discovery Sidebar (Filters)



Component Breakdown

VISUAL TYPE SELECTOR

Clear colour differentiator for the property types to reduce cognitive load.

LIST VIEW

Keeping the map as base as per the request the list view slides over the map.

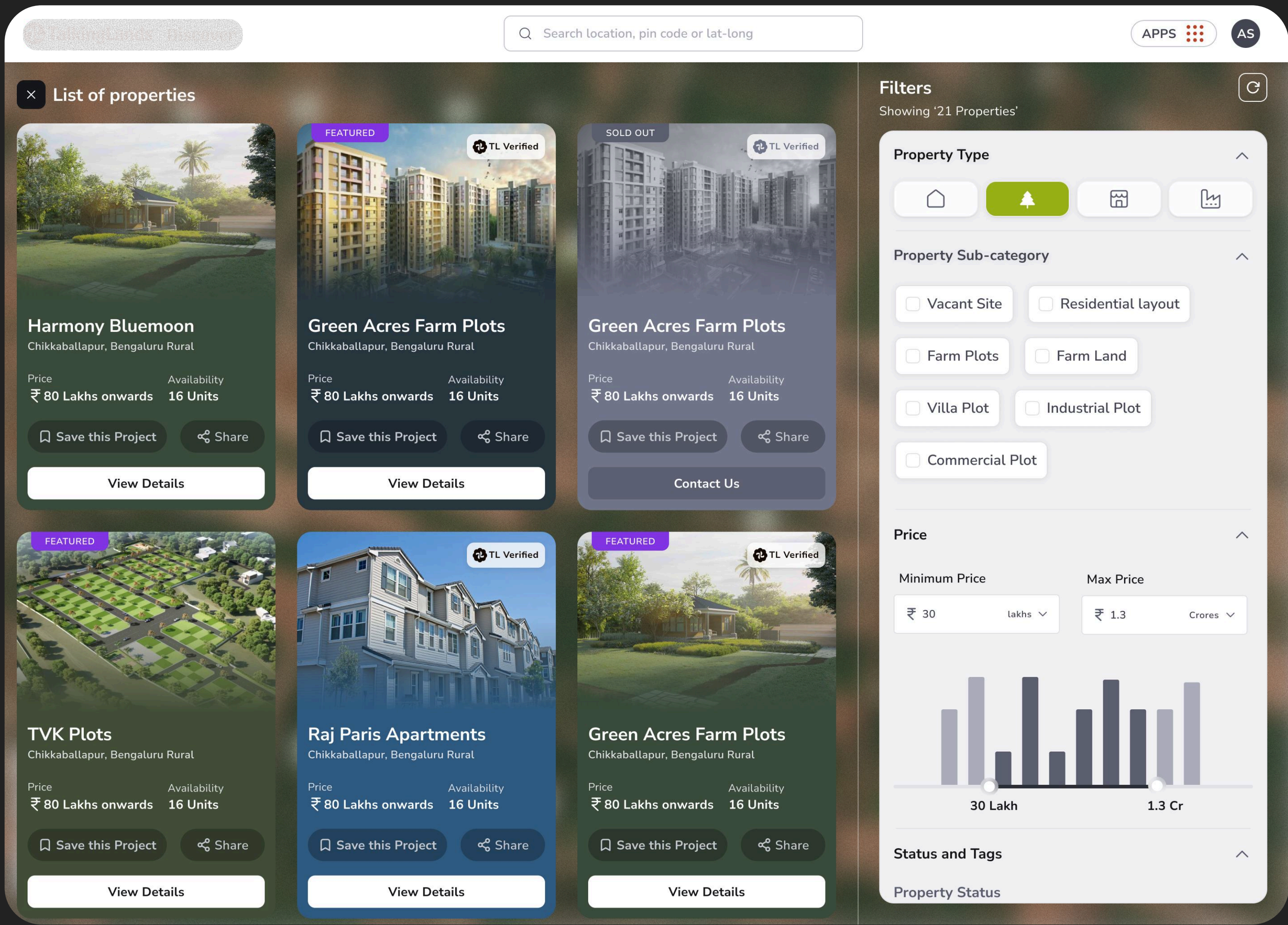
REAL-ESTATE USAGE

Fixed space to handle the filters, so that irrespective of the view filters stays same

DESIGN NOTE

Maintain airy spacing and ensure the sidebar background is neutral with the acrylic so that the map is always visible to provide clean contrast against the data-heavy map interface.

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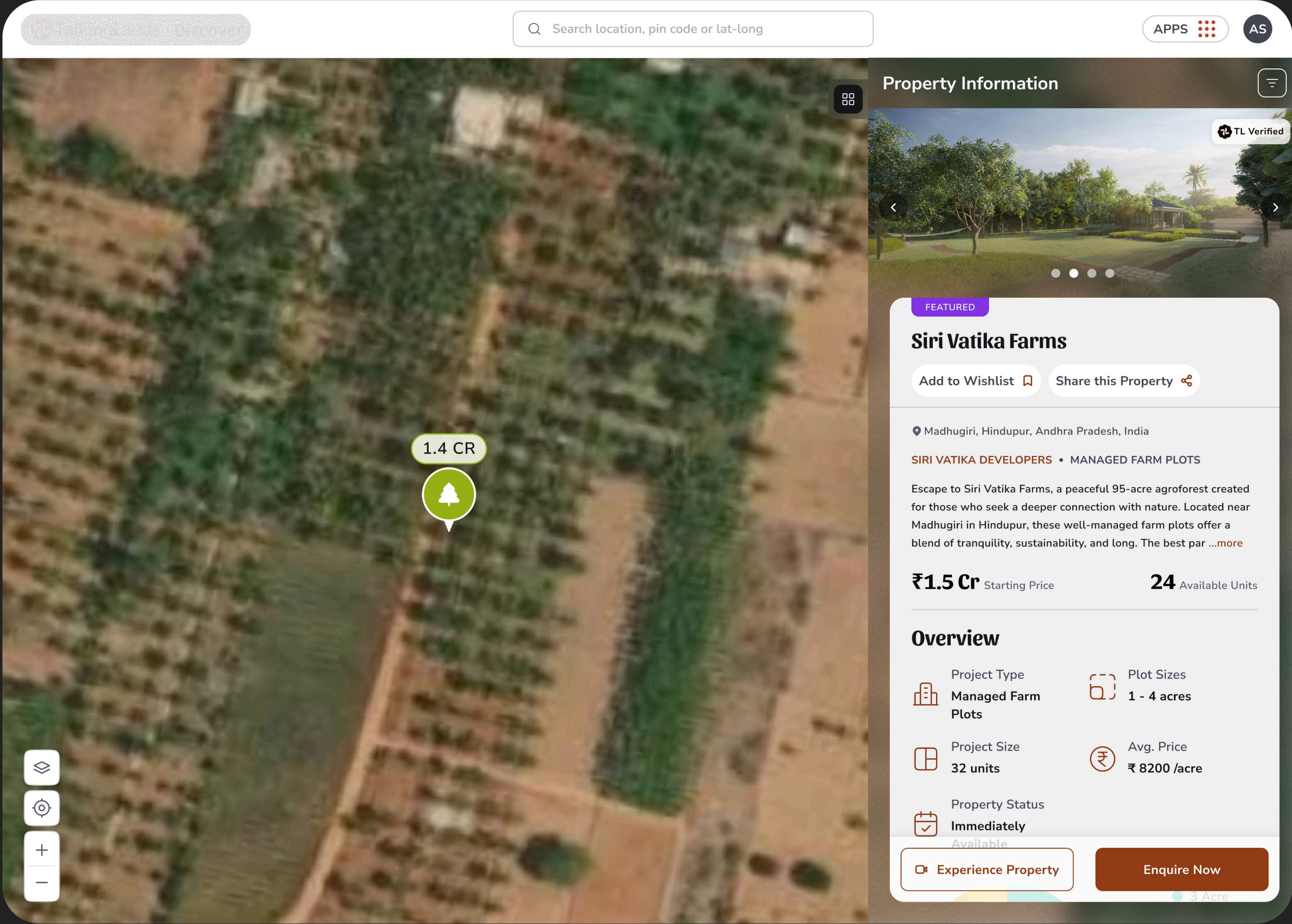
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Property Detail & Other features



Property Info

VISUAL TYPE SELECTOR

Visual consistency between the primary and secondary screens as user only zooms in to see property info

SCALABLE DATA

As per the property type the data varies based so a scalable info area with clear bucketing of data on a long scroll

COLOURS

Brand colour is minimal used as its data heavy application.

360 DEGREE VIEW

The 360 degree experience is redirected to a separate page where user can see all the media related docs in one place like a micro size for the developer

Mood Board Sample

Light theme

3 Insights

#8024F4 - App

#EAE7F9 - Neutral

Gradient

#16AF93 - App

#EDFAF8 - Neutral

Gradient

3 Objectives

#F46135 - App

#FAF1EF - Neutral

Gradient

Brand theme

#0052F2 - Brand

#1A1A1A - Brand

#FFFFFF - Brand

#0052F2 - Brand

#1A1A1A - Brand

#FFFFFF - Brand

Dark theme

#AC6CFA - App

#2F2B43 - Neutral

#37EFC3 - App

#202836 - Neutral

#FF6644 - App

#262120 - App

A color palette diagram for a pink/red theme. It features a central 'App' color (bright pink) with 'Highlights' (lighter pink) and 'Brand' (darker pink) variations above it. To the left, a 'Neutral' row shows a gradient from white to light pink, and a 'Grays' column shows a gradient from light gray to dark gray. To the right, a 'Tints' row shows a gradient from light pink to white, and a 'Shades' column shows a gradient from dark gray to black. The 'Dark' color is a very dark gray/black.

A color palette diagram for a purple theme. It features a central 'App' color (bright purple) with 'Highlights' (lighter purple) and 'Brand' (darker purple) variations above it. To the left, a 'Neutral' row shows a gradient from white to light purple, and a 'Grays' column shows a gradient from light gray to dark gray. To the right, a 'Tints' row shows a gradient from light purple to white, and a 'Shades' column shows a gradient from dark gray to black. The 'Dark' color is a very dark gray/black.

A color palette diagram for a blue theme. It features a central 'App' color (bright blue) with 'Highlights' (lighter blue) and 'Brand' (darker blue) variations above it. To the left, a 'Neutral' row shows a gradient from white to light blue, and a 'Grays' column shows a gradient from light gray to dark gray. To the right, a 'Tints' row shows a gradient from light blue to white, and a 'Shades' column shows a gradient from dark gray to black. The 'Dark' color is a very dark gray/black.

A color palette diagram for a teal theme. It features a central 'App' color (bright teal) with 'Highlights' (lighter teal) and 'Brand' (darker teal) variations above it. To the left, a 'Neutral' row shows a gradient from white to light teal, and a 'Grays' column shows a gradient from light gray to dark gray. To the right, a 'Tints' row shows a gradient from light teal to white, and a 'Shades' column shows a gradient from dark gray to black. The 'Dark' color is a very dark gray/black.

A color palette diagram for a green theme. It features a central 'App' color (bright green) with 'Highlights' (lighter green) and 'Brand' (darker green) variations above it. To the left, a 'Neutral' row shows a gradient from white to light green, and a 'Grays' column shows a gradient from light gray to dark gray. To the right, a 'Tints' row shows a gradient from light green to white, and a 'Shades' column shows a gradient from dark gray to black. The 'Dark' color is a very dark gray/black.

A color palette diagram for a yellow theme. It features a central 'App' color (bright yellow) with 'Highlights' (lighter yellow) and 'Brand' (darker yellow) variations above it. To the left, a 'Neutral' row shows a gradient from white to light yellow, and a 'Grays' column shows a gradient from light gray to dark gray. To the right, a 'Tints' row shows a gradient from light yellow to white, and a 'Shades' column shows a gradient from dark gray to black. The 'Dark' color is a very dark gray/black.

Design Deliveries

DESIGN STRATEGY & DELIVERIABLES

KEY UI COMPONENTS

High-Fidelity Screens

50+ screens covering discovery flows, property details, user journeys for 5 personas, and responsive layout states.

Component System

Reusable UI library with map markers, filter cards, property cards, and tags for consistent implementation.

AI build front-end

I designed and built the complete front-end architecture using modern AI-assisted development methodologies. This innovative approach minimized implementation complexity, requiring only minimal developer intervention (approximately 10% effort) for fine-tuning and edge cases. The application is production-ready and fully prepared for API integration and deployment

Team & Collaboration

Team Composition

- 1 Product Designer
(executing design direction)
- 4 Full-Stack Developers (implementing frontend and GIS integration)

Leadership Role

As Principal Design Consultant, I led strategic decision-making around information architecture, user flows, and design principles. Responsibilities included conducting user research, synthesizing insights, defining the spatial-first direction, and ensuring alignment between design and development throughout the delivery cycle.

Thank you

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